ISO 9001 BSI REGISTERED FM40574

Registration applies to the Boulder, Colorado facility.

HandiLaz Mini

Operator's Manual

www.pmeasuring.com

PARTICLE MEASURING SYSTEMS HEADQUARTERS
5475 Airport Blvd., Boulder, CO 80301, USA
(303) 443-7100 1-800-238-1801 FAX: (303) 449-6870
Instrument Service & Support: 1-800-557-6363
Customer Response Center: 1-877-475-3317

PARTICLE MEASURING SYSTEMS EUROPE
Tel: +44 (0)1684-581000 FAX: +44 (0)1684-560337
PMSEurope@pmeasuring.com

PARTICLE MEASURING SYSTEMS ASIA PACIFIC
Tel: (65) 8460-500 FAX: (65) 8460-700
PMSAsiaPacific@pmeasuring.com

PARTICLE MEASURING SYSTEMS, JAPAN
Tel: (81) 3-5298-8175 Fax: (81) 3-3255-8155
PMSJapan@pmeasuring.com

PARTICLE MEASURING SYSTEMS, CHINA
Tel: (86) 21-6113-3688 Ext. 751 Fax: (86) 21-6113-3683
PMSChina@pmeasuring.com

PARTICLE MEASURING SYSTEMS, MEXICO
Tel: 52 55 22 71 51 06
PMSMexico@pmeasuring.com

PARTICLE MEASURING SYSTEMS , PUERTO RICO
Tel: 787-380-5468 or 787-380-5446
PMSPuertoRico@pmeasuring.com

P/N 1000010551 Rev E

HandiLaz Mini Operator’s Manual
P/N 1000010551 Rev E

Quality Policy

The Quality Policy of Particle Measuring Systems is to strive to meet or exceed the needs and expectations of our customers and to align the activities of all employees with the common focus of customer satisfaction through continuous improvement in the quality of our products and services.

Environmental Information

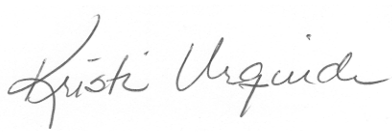
This equipment must be properly disposed of at end-of-life by means of an authorized waste management system. Information regarding dismantling of the equipment and location of any hazardous materials can be located on our web site at:
http://www.pmeasuring.com/support/papers/disassemblyinstructions.

CE - Declaration of Conformity

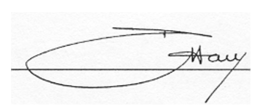
Application of Council Directive(s):	89 / 336 /EEC, 73 / 23 /EEC	
Standard(s) to which Conformity is Declared:	EMC	EN 61326: 1997/A1: 1998/A2: 2001
	Safety	EN 61010-1: 2001
Manufacturer's Name:	Particle Measuring Systems, Inc.	
Manufacturer's Address:	5475 Airport Boulevard, Boulder, CO 80301 USA	
Manufacturer's Telephone/FAX:	(303) 443-7100 / (303) 449-6870	
Distributor's Name:	Particle Measuring Technique, Great Britain	
Distributor's Address	Unit B1 South, Willow End Park	
	Danemoor, Malvern WR14 1XZ	
	Worchestershire, WR13 6NN United Kingdom	
Distributor's Telephone/FAX:	011-44-1684-312950 / 011-44-1684-312969	
Type of Equipment:	Particle Counter	
Model No:	HandiLaz Mini	

I, the undersigned, hereby declare that the equipment specified above conforms to the above Directive(s) and Standard(s).

Signature:



Signature:



Full Name:	Kristi Urquidi	**Full Name:**	David Hall
Position:	Vice President Operations	**Position:**	Managing Director
Place: Boulder	Date: December 6, 2005	**Place: Malvern**	Date: December 6, 2005

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Chapter 1
Introduction

The HandiLaz® Mini is a handheld, 3-channel, airborne-particle counter designed for speedy assessment of microcontamination levels in cleanrooms and other controlled environments. Its molded chassis is designed for ease of handling and cleaning. It can be powered either from a direct AC source, or from four internal NiMH batteries that can be recharged.

The HandiLaz Mini is ergonomically designed to fit in the palm of your hand. The Mini is rugged and dependable, providing tremendous value for cost-conscious users who want a reliable, easy-to-use, handheld particle counter.

Features:

- Small and light enough to be carried comfortably for long periods
- Simple to use and maintain
- Sizing sensitivities from 0.3 to 5.0 microns
- Sample flow of 0.1 ft.^{3}/min. (2.83 LPM)
- The internal batteries can be charged rapidly
- Stores data internally for review, print, or transfer to a PC
- Interfaces with an external PC to allow result analysis (using software)
- Optional external printer

Control Keys

ENTER	• Moves to the highlighted screen **or** • Enters the selected option, and then moves to the next choice.
PREV	Returns to *previous* screen.
UP/DOWN Arrows	• Changes displayed value **or** • Toggles between screen displays (e.g., from 0.5 to 5.0 µm, or from #/Ft.^{3} to #/M^3) **or** • Moves to the next stored record
START/STOP	• Starts or Stops sampling **or** • Starts display or printing of records

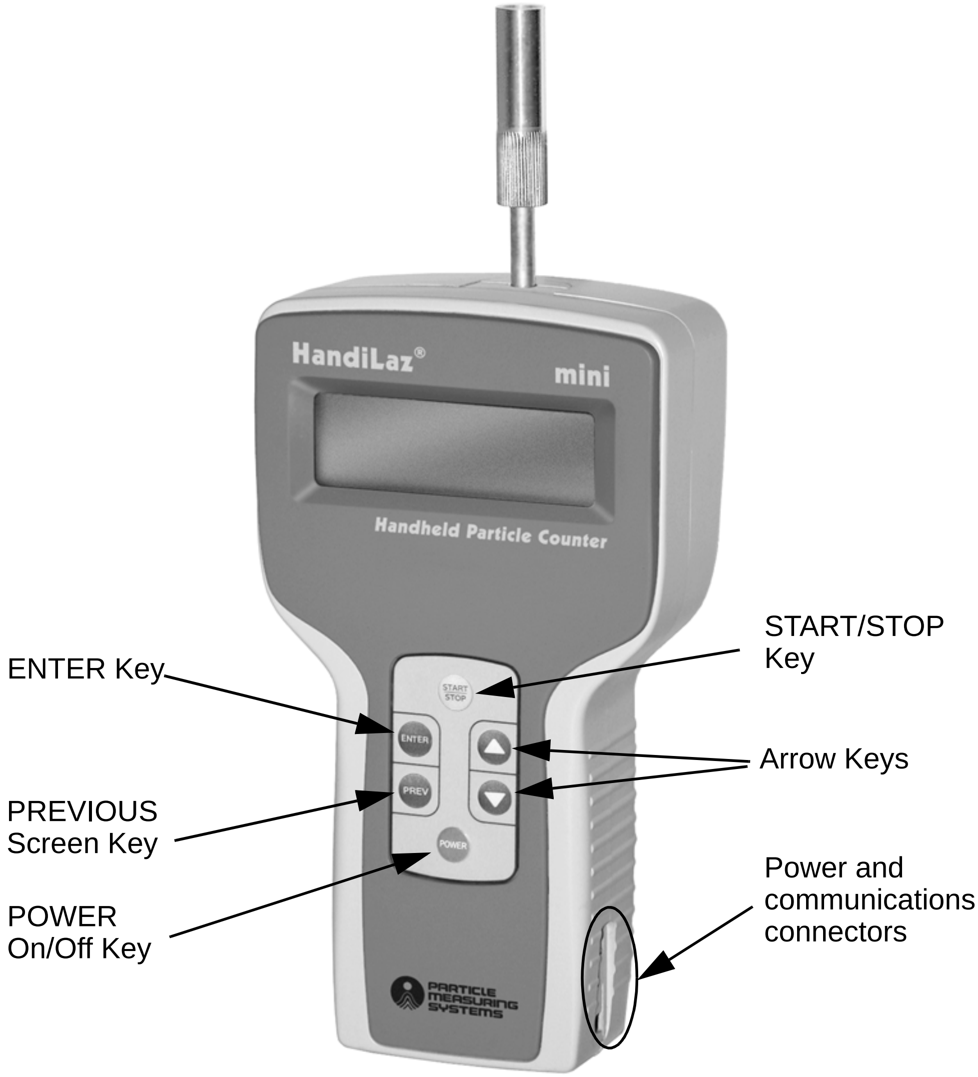

Figure 1-1: HandiLaz Mini

Specifications

Channel thresholds:	0.3, 0.5, and 5.0 µm
Flow rate:	0.1 cfm (2.83 liters per minute)
Flow sensor:	Smart flow control based on pressure transducers
Light source:	Laser diode
Calibration particles:	NIST-traceable particles
Maximum concentration (@ 5% coincidence error):	>2,000,000/ft.^{3}
Zero count:	<1 per five min. (meets JIS)
Counting efficiency:	50% ± 20% @ 0.3 µm (meets JIS for first channel)
ISO Classes:	Best for ISO 5–8
Sample output filtering:	None
Communication modes:	RS-232C or RS-485
Communications protocol:	9600 bps Baud rate
Data display modes:	Counts, Counts/m^3, Counts/ft.^{3}
Sampling modes:	• Single • Repeat • Continuous • Calculation • ISO (calculations for 0.5 and 5.0 µm)
Key internal software features:	• ISO certification, Hotkey, Autostart, Sampling, Display, Uplink, Print, Clear and Alarms
Standard reports (from optional external printer):	Repeat, Single or Continuous mode, Calculation or ISO mode
Error messages (real-time):	• Excessive Concentration • Exceeds Alarm Limits • Laser Power • Low Battery, Printer Buffer
External software:	Downloads CSV file
Data storage:	Up to 10,000 samples. Calculations and ISO modes take more than one storage space

Data security:	Cannot edit stored data
External surface:	ABS (plastic resin)
Appropriate cleaning materials:	Methanol or neutral detergents
Dimensions (w, h, d):	4.3 × 7.6 × 2.7 inches (11 × 20 × 7 cm)
Weight:	1.5 pounds (0.68 kg) without batteries
Power:	AC adapter: 100–240 VAC, 50/60 Hz
Batteries:	4 rechargeable, AA NiMH batteries (1.2 VDC, 2100 mA)
Operating time on battery:	3 hours
Battery charging:	External; 3.5 hours
Operating environment:	• Temperature: 10-35° C • Humidity: 0–85%, non-condensing • Area type: cleanroom

NOTE: Specifications are subject to change without notice.

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Chapter 2
Taking Samples

Introduction

While the HandiLaz Mini offers 5 different sampling modes, the setup and screens are quite similar for all. In fact, if you want to use the same setup you used last time, it takes as few as 6 key pushes to start sampling. If this isn't fast enough, you can use the HOTKEY option (see Chapter 5) to reduce this to only two key pushes. Sampling setup screens are quite similar from one sampling mode to another, using a version of the same setup screen. (See Chapter 3 for more details on the different sampling modes.)

Sample in Six Key Pushes (Using current sampling settings)

1. Press Power = ON.
2. <MENU> will be displayed, with the cursor on "MEASURE MODE."
   Press ENTER.
3. <MEASURE MODE> will be displayed, with the cursor on "REPEAT."
   Press ENTER.
4. REPEAT Screen will be displayed, with the cursor on "NO." Press the Arrow key to change value to "OK."
5. Press ENTER.
6. SAMPLING Screen will be displayed, with the cursor on "OK." Press ENTER.

Sample in Two Key Pushes Using HOTKEY (Using designated mode & settings)

1. Press Power = ON.
2. Press START/STOP.

Sampling Sequence

If you have preset the unit to AUTOSTART, the unit will conduct sampling as follows:

1 "WAIT" will be displayed for preset period (min. of 10 sec.), allowing the unit to calculate and stabilize the flow, plus the allowing the operator to withdraw from the area near the sampling probe.

2 "RUN" will be displayed while the unit takes the first sample.

3 "WAIT" will be displayed for the inter-sample interval, i.e., between sequential samples from the same location. For sampling to be skipped for an inter-sample interval of M seconds, the Cycle Time (i.e., "INT") must be set M seconds longer than the "SAMPLE" time.

4 "RUN" will be displayed while the unit takes the next sample.

5 "END" will be displayed when the sampling sequence is completed.

If you have NOT preset the unit to AUTOSTART, after the "WAIT" period is completed, the unit will display "READY." Press the START/STOP key to begin sampling and display "RUN."

Changing Sampling Settings

The sampling setup screen is shown in Figure 2-1. The key parameters are:

Sampling Mode Already selected by user in <MEASURING MODE>

Particle Size Channels Automatically set by selecting the Sampling Mode. (ISO Mode only uses 0.5 and 5.0 µm channels)

Location Number Tags data record and print strip with a numeric ID. (Allows values from 000 to 255)

STR Stores data in unit if set = Y (for YES)

BEEP Provides audible beep is set = Y (for YES)

SAMPLE Time Duration of single sample (in MM:SS)

TIMES Number of sequential samples taken at this location

INT Cycle Interval = Time to take sample + Inter-sample Delay
(If INT set = Sample time, then Inter-sample Delay = 0.
Once unit notices, it will reset the lower value to = Sample time)

NO Sampling setup is NOT completed

OK Sampling setup IS completed, ready to start sampling

Figure 2-1: Sampling Setup Screen: REPEAT Mode

Data Display

At the end of sampling the results are displayed on the LED screen (see Figure 2-2). The displayed information is as follows:

Sampling Mode Mode used to collect data

Storage Location Last Saved Sample is stored in this location

Beep Mode Status B shows Beep Mode = YES;
"blank" shows Beep Mode = NO

Battery Icon Icon outline with blank interior = Battery is very low.
Icon blacked-in = Battery has adequate charge.

END Sampling sequence has been completed

0.3, 0.5 & 5.0 Channel size labels: Cumulative counts of that size or larger

CNT Units set to show as raw Counts (alternatively, choose #/Ft^3 or #/M^3)

 # Samples Completed / # Samples Scheduled Example: "02/03" means the program completed 2 samples
at this location, compared to the 3 samples scheduled.

:

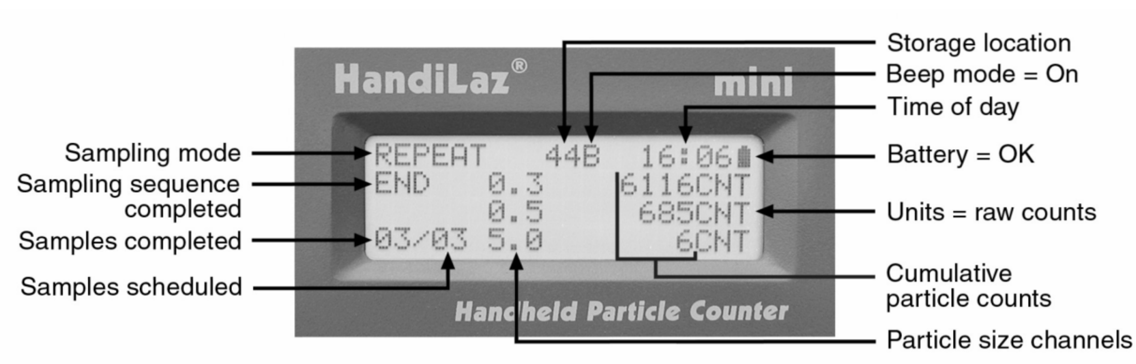

Figure 2-2: Data Display Screen: REPEAT Mode

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Chapter 3
Measure Mode Settings

This chapter describes the menu items on the MEASURE MODE screen.

```
     <MEASURE MODE>
   1.REPEAT   4.CALC
   2.SINGLE   5.REMOTE
   3.CONT     6.ISO
```

REPEAT Use this sampling mode to take a series of samples from the same location.

SINGLE Use this sampling mode to take a single sample from a location.

CONT (i.e., Continuous) Use this sampling mode to continuously sample until the operator stops the HandiLaz Mini by pressing the START/STOP key.

CALC (i.e., Calculate) Use this sampling mode to take a preset number of samples, then calculate the Average, Standard Deviation, Minimum, and Maximum values for each size channel.

REMOTE This mode is not supported.

ISO Use this mode to sample and perform ISO calculations. Because of the greater time required to sample in ultraclean rooms, this 0.1 CFM sample flow instrument is the most time-efficient in certifying rooms of ISO Class 5 through 8. The ISO measurement mode considers 0.5 and 5.0 µm particles only.

Using REPEAT Sampling Mode

Use this sampling mode when several sequential samples are needed from the same location.

To open the REPEAT sampling mode:

1. Use either arrow key to navigate to MEASURE MODE on the main menu.
2. Press the ENTER key.

   The <MEASURE MODE> menu will open.
3. Use either arrow key to navigate to REPEAT.

4 Press the ENTER key.

The REPEAT measure mode setup screen will open.

```
REPEAT 0.3/0.5/5.0µm
LOC.023 STR:N BEEP:N
SAMPLE 00:15  3TIMES
INT 00:00:30      NO
```

NOTE: The **NO** in the lower-right corner indicates that the setup is not complete and the screen is ready to be edited. If setup is complete, set **NO** to **OK**.

REPEAT Sample Parameters

The parameters displayed when the REPEAT measure mode is opened are those that were saved when the REPEAT mode was last run. The REPEAT mode can be run as it is, without changing any sample parameters.

To edit the setup screen:

1 Press the ENTER key.
Pressing ENTER moves the cursor in a loop through each field that can be edited.

2 At the field to be changed, press the up or down keys.
The field setting will loop through its range of allowable settings.

3 When the field is set correctly, press the ENTER key again to move to the next field.

To close the edit function and run the sample:

1 Use the ENTER key to navigate to NO.

2 Use the up or down key to toggle the NO to OK.

3 Press the START/STOP key.

The SAMPLING display will appear.

```
     SAMPLING
SAMPLE TIME   00:00:45
TOTAL TIME    00:01:15
FREE REC.:09500    OK
```

Taking REPEAT Samples

From the SAMPLING display, press the ENTER key. During the initial delay, the HandiLaz Mini displays WAIT. When READY is displayed, press START/STOP to begin sampling. Alternatively, the unit can be set to AUTOSTART sampling. See “Setting AUTOSTART (AUTOST)” on page 5-6.

SINGLE Sampling Mode

Use this sampling mode when taking only one sample.

To open the SINGLE sampling mode:

1. Use either arrow key to navigate to MEASURE MODE on the main menu.
2. Press the ENTER key.

   The MEASURE MODE menu will open.
3. Use either arrow key to navigate to SINGLE.
4. Press the ENTER key.

   The SINGLE measure mode setup screen will open.

```
SINGLE 0.3/0.5/5.0µm
LOC.023 STR:N BEEP:N
SAMPLE 00:15
FREE REC.:09950    NO
```

Setting SINGLE Sample Parameters

The parameters seen when the SINGLE measure mode is opened are those that were saved when the repeat mode was last run. The SINGLE measuring mode can be run as it is, without changing any sample parameters.

To edit the setup screen:

1. Press the ENTER key.

   Pressing the ENTER key moves the cursor in a loop through each field that can be edited.
2. At the field to be changed, press the up ▲ or down ▼ key.
   The field setting will loop through its range of allowable settings.

To close the edit function and run the sample:

1. Use the ENTER key to navigate to NO at the lower-right corner of the display.
2. Use the up ▲ or down ▼ key to toggle the NO to OK.
3. Press the ENTER key.

   The SAMPLING display will appear.

```
     SAMPLING
SAMPLE TIME   00:00:45
TOTAL TIME    00:01:15
FREE REC.:09500    OK
```

Taking SINGLE Sample

1. From the SAMPLING display, press the START/STOP key.
   The initial delay countdown will begin.
2. After the initial delay, press the START/STOP key.

Using CONT (Continuous) Sampling Mode

Use this sampling mode to continuously sample until the operator stops the HandiLaz Mini by pressing the START/STOP key. The maximum duration of a continuous sample is 99 minutes and 59 seconds.

To open the CONT sampling mode:

1. From the main menu, use either arrow (▲ or ▼) key to navigate to MEASURE MODE.
2. Press the ENTER key.

   The MEASURE MODE menu will open.
3. Use either arrow key to navigate to CONT.
4. Press the ENTER key.

   The CONT measure mode setup screen will open.

```
SINGLE 0.3/0.5/5.0µm
LOC.023 STR:N BEEP:N

FREE REC.:09950    NO
```

Setting CONT (Continuous) Sample Parameters

The parameters seen when the CONT measure mode is opened are those that were saved when the CONT mode was last run. The CONT measuring mode can be run as it is, without changing any sample parameters.

To edit the setup screen:

1. Press the ENTER key.

   Pressing the ENTER key moves the cursor in a loop through each field that can be edited.
2. At the field to be changed, press the up ▲ or down ▼ key.
   The field setting will loop through its range of allowable settings.
3. When the field is set correctly, press the ENTER key again to move to the next field that can be edited.

To close the edit function and run the sample:

1 Use the ENTER key to navigate to NO.

2 Use the up or down key to toggle the NO to OK.

3 Press the ENTER key.

The SAMPLING display will appear.

```
     SAMPLING
SAMPLE TIME   00:00:45
TOTAL TIME    00:01:15
FREE REC.:09500    OK
```

Taking CONT (Continuous) Sample

Press the ENTER key. After the initial delay, press STOP/START to begin sampling. Sampling continues until the operator stops it.

CALC (Calculation) Mode

Use this sampling mode to have the HandiLaz Mini take the preset number of samples, calculate the Average, Standard Deviation, Minimum, and Maximum values for each size channel.

To open the CALC sampling mode:

1 Use either arrow key (▲ or ▼) to navigate to 1.MEASURE MODE on the main menu.

2 Press the ENTER key.
The MEASURE MODE menu will open.

3 Use either arrow key to navigate to 4.CALC.

4 Press the ENTER key.

The CALC measure mode setup screen will open.

```
CALC    0.3/0.5/5.0µm
LOC.023 STR:N BEEP:N
SAMPLE 00:15  3TIMES
FREE REC.:09500   NO
```

Setting CALC (Calculation) Sample Parameters

The parameters seen when the CALC measure mode is opened are those that were saved when the CALC mode was last run. The CALC measuring mode can be run as it is, without changing any sample parameters.

To edit the setup screen:

1. Press the ENTER key to move the cursor in a loop through each field that can be edited.
2. At the field to be changed, press the up ▲ or down ▼ key.
   The field setting will loop through its range of allowable settings.
3. When the field is set correctly, press the ENTER key again to move to the next field that can be edited.

To close the edit function and run the sample:

1. Use the ENTER key to navigate to NO.
2. Use the up or down key to toggle the NO to OK.
3. Press the ENTER key.
   The SAMPLING display will appear.

```
     SAMPLING
SAMPLE TIME   00:00:45
TOTAL TIME    00:01:15
FREE REC.:09500    OK
```

Taking CALC (Calculation) Sample

From the SAMPLING display, press the ENTER key. After the initial delay, press STOP/START to begin sampling.

ISO Sampling Mode

Use this mode to sample and perform ISO calculations. Because of the greater time required to sample in ultraclean rooms, this 0.1 CFM sample flow instrument is the most time-efficient in certifying rooms of ISO Class 5 through 8. The ISO measurement mode considers only 0.5 and 5.0 µm particles.

To open the ISO sampling mode:

1. From the main menu, use either arrow key to navigate to MEASURE MODE.
2. Press the ENTER key.
   The MEASURE MODE menu will open.

3 Use either arrow key to navigate to **ISO**.

4 Press the ENTER key.

The ISO sampling mode setup screen will open.

```
ISO      0.5/5.0µm
        STR:N BEEP:N
SAMPLE 00:15   3TIMES
INT 00:00:15       OK
```

Setting ISO Sample Parameters

The parameters seen when the ISO measure mode is opened are those that were saved when that mode was last run. The ISO measuring mode can be run as it is, without changing any sample parameters.

To edit the setup screen:

1 Press the ENTER key to move the cursor in a loop through each field that can be edited.

2 When the cursor is over the field to be changed, press the up ▲ or down ▼ key. The field setting will loop through its range of allowable settings.

3 When the field is set correctly, press the ENTER key again to move to the next field that can be edited.

To close the edit function and run the sample:

1 Use the ENTER key to navigate to **NO** at the lower-right corner of the display.

2 Use the up ▲ or down ▼ key to toggle the **NO** to **OK**.

3 Press the ENTER key.

The SAMPLING display will appear.

```
     SAMPLING
SAMPLE TIME   00:00:45
TOTAL TIME    00:01:15
FREE REC.:09500    OK
```

Taking ISO Sample

From the sampling display, press the ENTER key. After the initial delay, press START/STOP to conduct sampling at the first location.

ISO sampling typically requires sampling from multiple locations. Once the sampling is completed at the first location, the query “NEXT POINT?” will be posted. At this time the user presses arrow keys to select one of the following alternatives:

NEXT POINT? Pressing ENTER starts the unit sampling at the next location, taking the same number of samples as at the current location.

FINISH? Pressing ENTER concludes sampling, makes all ISO calculations, and (if STR=Y) saves the ISO samples and calculations to memory.

CALCULATE? Pressing ENTER calculates the Average, Standard Deviation, and 95% Upper Confidence Limit for the samples already collected, and displays the results for the 0.5 µm particles. Pressing ENTER again toggles to display the results for 5.0 µm particles. Pressing **PREV** will return to “NEXT POINT?”.

EDIT TIMES? Pressing ENTER allows the user to change the number of samples to be taken at the next location.

ERASE DATA? Pressing ENTER erases the data from the most recent location sampled.

Chapter 4
Data Processing Settings

This chapter includes all the functions under the menu item “DATA Processing”.

- DISPLAY
- UPLINK
- PRINT
- CLEAR

DISPLAY

This function allows the user to review the records stored in the HandiLaz Mini.

NOTE: The HandiLaz Mini only stores records when STR (store) has been set to Y (yes) ***before*** sampling.

To open the DISPLAY function:

1. From the main menu, use the up ▲ or down ▼ keys to navigate to DATA PROCESSING
2. Press the ENTER key.
   The DATA PROCESSING menu will be displayed.
3. Use the up or down keys to navigate to DISPLAY.
4. Press the ENTER key.
   The following display will appear:

```
     &lt;DISPLAY&gt;
RECORDS  00050
  START 00001
```

The HandiLaz Mini can store up to 10,000 records in memory. This display indicates that the unit has 50 records stored in its memory, and that, if nothing is changed, the first record viewed will be record 00001.

To select which record will be viewed first:

1 Press the ENTER key until the cursor moves to the digit position to be modified.

2 Press the up ▲ or down ▼ keys as needed to change the digit(s) to specify the record which will be reviewed first.

Example – To view the 13th record first:

1 Press the ENTER key until the cursor moves over the second digit from the right.

2 Press the up key ▲, changing the zero to 1.

3 Press the ENTER key again to move the cursor over the right-most digit.

4 Press the up ▲ key until the 1 becomes 3.

5 Press the START/STOP key to open the 13th record.

NOTE: In this example, trying to change any of the first three digits on the left will cause the HandiLaz Mini to generate three rapid "beeps", indicating an error. The digits in those leading positions cannot be changed because there are only 50 records. If 150 records were stored, the last two digits on the left would be insignificant and therefore unchangeable.

To view records in sequence:

1 From any start position number, press the START/STOP key. The record number indicated in the START line will open.

2 Press the up ▲ key to see the next higher record in order.

NOTE: Holding the up or down key down causes the records to open in sequence at about two records per second.

Display Example:

Location

```
00001      9/06  09:51
0.3µm   5.08E+5/cf
0.5µm   4.14E+4/cf
5.0µm   4.00E+1/cf
```

Date, Time
Number of particles ≥ 0.3 µm
Number of particles ≥ 0.5 µm
Number of particles ≥ 5.0 µm

CALC (Calculation) Records

In CALC mode, the samples are not saved, but the calculated results are. Saving the CALC results takes two storage spaces.

Records comprising a CALC mode sample are preceded by a message similar to the following:

```
RECORDS 011-012>
CALCULATION MODE
  09/13  09:35
  LOCATION 001
```

UPLINK

The UPLINK function is used with PC Link software (also referred to as the Data Retrieval Software) to copy data from the HandiLaz Mini to a PC. See Chapter 6, "Downloading Data to PC".

The data ***cannot*** be retrieved unless the particle counter is first placed in UPLINK mode. If the Data Retrieval Software attempts to download the data and the unit is ***not*** in UPLINK mode, the PC will display an “RS-232 Error.”

To put the unit in UPLINK mode:

1. Connect the communications cable from the HandiLaz Mini to the PC
2. From the main menu Select DATA PROCESSING and then press the ENTER key.
3. Select UPLINK and then press the ENTER key.

PRINT

This function prints one or more of the data records on the optional external printer. The user can choose to print the following:

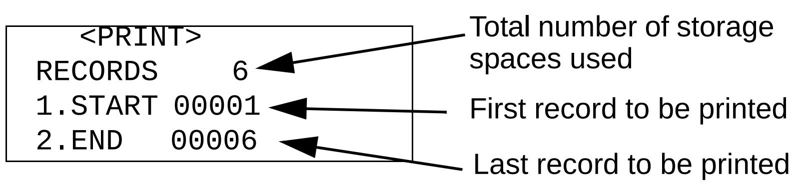
- All records
- A range of records—the user selects the first and last record to be printed. All intermediate records will also be printed.
- A single record

NOTE: If the printer has not been used before, its mode settings and serial settings must be configured. See Chapter C, "Printer Setup".

To print from the HandiLaz Mini:

1. Connect the printer cable from the HandiLaz Mini to the printer.
2. Ensure that HandiLaz Mini communications setting is RS232C.
3. From the main MENU select DATA PROCESSING and press the ENTER key.
4. In the <DATA PROCESSING> menu, navigate to PRINT and press the ENTER key.

   The following display will appear:



Example – Print records 2, 3, and 4:

1. From the <PRINT> display seen above, press the ENTER key until the cursor is over the 1.
2. Use the down ▼ key to change the 1 to 2.
3. Press the ENTER key again, until the cursor moves over the 6.
4. Press the down ▼ key until the 6 changes to 4.

   The <PRINT> function is now set to print records 2 through 4.
5. Press the START/STOP key. The printer will begin printing and stop when all records in the range are printed.

To print a single record:

1 Follow the instructions for printing a range of records (above, but set START and END to the same record number that is to be printed.

2 Press the START/STOP key.

CLEAR

The CLEAR function deletes **all** the sample data stored in the HandiLaz Mini.

To clear all the sample data stored in the HandiLaz Mini:

1 From <MENU> select DATA PROCESSING and press ENTER.

2 In the <DATA PROCESSING> menu, navigate to CLEAR and press ENTER.

The following display will appear:

```
  <DATA CLEAR>
 RECORDS 00053
   CLEAR  NO
```

The only user-option is to change CLEAR from NO to YES. YES will delete ALL records.

3 Use the up ▲ or down ▼ key to set CLEAR to YES.

The following display will appear:

```
  <DATA CLEAR>

  ARE YOU SURE?
ENTER:YES  PREV:NO
```

4 To delete ***all*** the data, press the ENTER key. To escape the DATA CLEAR process, press the PREV key.

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Chapter 5
Options Settings

The OPTIONS settings include the following:

- ALARM
- UNITS
- UTILITIES
  - ➢ DATE
  - ➢ TIME
  - ➢ ID (instrument)
  - ➢ Communications Protocol
- HOTKEY
- AUTOST (i.e., AUTOSTART)
  - ➢ AUTOSTART: Setting Initial DELAY
  - ➢ AUTOSTART: ON/OFF

Setting ALARM

HandiLaz Mini can be set to generate an alarm when particle concentration exceeds a user-set level for the particle size channels. When the concentration level is exceeded in a size channel, the channel size label (0.3, 0.5, and 5.0) will begin blinking. One or more asterisks will be displayed beside the channel label on both the display and the printout.

To change the ALARM settings:

1. Use the up ▲ or down ▼ key to navigate to OPTIONS on the main menu
2. Press the ENTER key.
   The OPTIONS menu will open.

3 If necessary, use the up or down key to navigate to the ALARM menu item and then press the ENTER key.
The ALARM edit display will open.

```
      <MEASURE MODE>
1. 0.3µ   7.00E+7
2. 0.5µ   1.00E+3
3. 5.0µ   1.00E+2
```

About Scientific Notation: The notation of 7.00+7 is another way expressing 7.00 times 10 to the 7th power, or 70,000,000.

4 Press the ENTER key as needed to navigate to the value to be edited.

5 Press the up ▲ or down ▼ key to change the value as desired.

6 When all the values have been set, press the PREV key to exit the ALARM display.

During sampling, when the particle count exceeds the user-set maximum concentration or count, the label for the particle size on the display will begin to blink.

Setting UNITS

Particle data can be displayed and stored in the following units:

Raw particle count	CNT
Counts per cubic foot	cf
Counts per cubic meter	m3

Changing the units effects only the sample data collected ***after*** the change of units. Data that has already been collected and stored can not be changed.

To change the units in which data is displayed and saved:

1 Navigate to OPTIONS on the main menu.

2 Press the ENTER key. The OPTIONS menu will open.

3 Navigate to UNITS, and press the ENTER key.
The UNITS display will open.

```
  <UNITS>
  SETUP
1.PTCL   CNT
```

4 Press the up ▲ or down ▼ key to toggle through the available units settings.

5 When the desired units setting is displayed, press the PREV key to exit the UNITS SETUP display.

Setting UTILITIES

The items set in Utilities are the following:

- Date
- Time
- Instrument ID
- Communications Protocol

Setting Date, Time, or Instrument ID

The date and time can be set whenever necessary. Changing the date and time will effect only those samples collected after the change. Stored data will be unaffected. An internal Lithium battery retains the date, time, and ID when the AA batteries are removed.

About the Instrument ID The Instrument ID is the unique address the PC-LINK software must use to communicate with the HandiLaz Mini. No other HandiLaz Mini should be assigned the same number.

To change the date, time, or instrument ID:

1. Navigate to OPTIONS on the main menu.
2. Press the ENTER key. The OPTIONS menu will open.
3. Navigate to UTILITIES, and press the ENTER key.
   The UTILITIES setup display will open.

```
      <UTILITIES>
1.DATE     2005/10/10
2.TIME     09:46
3.ID      03 (MAX 31)
```

4. Press the ENTER key to navigate to the field to be changed.
5. Press the up ▲ or down ▼ key until the field displays the correct value.
6. Repeat steps 4 and 5 until all fields are correct.
7. Press PREV to exit the UTILITIES setup display.

NOTE: Pressing the ENTER key while the second digit of the ID is active opens the Communications Protocol menu.

Setting Communications Protocol

The HandiLaz Mini can be set to communicate with a PC by means of the following communications protocols:

- RS232C or
- RS485

To set the Communications protocol:

1. If the <UTILITIES> screen is not visible, complete the following 3 steps (step 2–4). Otherwise, go to step 5, below.
2. Navigate to OPTIONS on the main menu.
3. Press the ENTER key. The OPTIONS menu will open.
4. Navigate to UTILITIES, and press the ENTER key.
   The UTILITIES setup display will open.

```
      <UTILITIES>
1.DATE     2005/10/10
2.TIME     09:46
3.ID       03 (MAX 31)
```

5. Press the ENTER key until the following screen is displayed.

```
      <UTILITIES>
4.COMMUNICATION
             RS232c
```

or RS485

6. Press the up ▲ or down ▼ key to toggle the setting from RS232C to RS485.
7. When the setting is correct, press the PREV key to exit the UTILITIES setup screen.

Setting HOTKEY

HOTKEY enables a user to start sampling with a single key push. When HOTKEY is enabled, and the unit is in the MAIN or MEASURE Mode Screen, then pressing the START/STOP key (i.e., the HOTKEY) automatically starts the sampling process. It begins with the countdown of the initial delay and operates in the designated sampling mode, according to the current settings.

To set the HOTKEY:

1. Navigate to OPTIONS on the main menu.
2. Press the ENTER key.
   The OPTIONS menu will open.

3 Navigate to UTILITIES, and press the ENTER key.
The UTILITIES setup screen will open.

4 Navigate to HOTKEY, and press the ENTER key.
The HOTKEY setup display will open.

```
<HOTKEY>
 REPEAT
```

Example: The HOTKEY setting seen above is “REPEAT”. This means that whenever the <MENU> screen is visible, pressing the START/STOP key will start sampling using the REPEAT mode’s sample parameters.

5 Press the up ▲ or down ▼ key until the desired sampling mode is displayed.

6 Press the PREV key to exit the HOTKEY setup display.

NOTE: Ensure that the selected MEASURING MODE has the desired sample settings.

Taking Samples Using HOTKEY

To run a sample with the HOTKEY:

1 Start the HandiLaz Mini. The <MENU> will display.

2 Press the START/STOP key.
The pump will immediately start, and the sampling display will open with the name of the pre-selected HOTKEY Measuring Mode appearing on the left of the display.
When the initial delay expires, the HandiLaz Mini will do one of the following, depending on its settings:

- Begin sampling immediately if AUTOST (Autostart) is set to ON.
- Enter the READY state, requiring the user to press the START/STOP key to begin sampling.

Setting AUTOSTART (AUTOST)

Access the <AUTOSTART> setup screen to set the initial delay or the AUTOSTART ON/OFF option.

Initial DELAY

When a sample is started, a user-set *initial* delay occurs before HandiLaz Mini begins counting particles. This delay allows the HandiLaz Mini to stabilize sample flow and check other internal functions. The minimum delay is 10 seconds, but the user can set the delay to any value up to 99 seconds.

AUTOSTART

HandiLaz Mini can be set to automatically start sampling after the initial delay expires. This eliminates the need to manually start sampling after the READY message is displayed.

In other words: after the initial delay, the HandiLaz Mini will do one of the following things:

- When AUTOSTART is ON, HandiLaz Mini will start counting particles without requiring any operator action.
- When AUTOSTART is OFF, HandiLaz Mini will enter the READY state, requiring the operator to press the START/STOP key to start sampling.

To set DELAY and AUTOSTART:

1. From the <MENU> screen, select OPTIONS.
2. In the <OPTIONS> menu screen, select AUTOST.
   The <AUTOSTART> setup screen will open.

```
   <AUTOSTART>
     DELAY: nnSEC
AUTOSTART: OFF
```

3. Use the ENTER key to navigate to the Initial DELAY fields.
4. Use the up ▲ or down ▼ key as necessary to set both fields.
5. Press the ENTER key to go to the AUTOSTART ON/OFF setting.
6. Use the up ▲ or down ▼ key to set AUTOSTART ON/OFF.
7. Press the PREV key to exit the AUTOSTART setting screen.

Chapter 6
Downloading Data to PC

Introduction

The PC Link Software provided with the HandiLaz Mini can be used to download data to a PC. Once in the PC, the data may be:

- Displayed for review
- Saved as a CSV file
- Printed in pre-defined report format on your standard PC or network printer.

Installing PC Link Software:

1. Turn off other applications.
2. Place the installation CD in the CD-ROM drive, then double-click the installation icon. The program steps the user through the complete installation process. Unless otherwise specified, the software will be saved in a newly-created directory called "PMS."
3. Enter the PMS directory (C:\Program Files\PMS) and double-click on the file entitled "Handheld LPC Data Retrieval Software Operating Manual." Once the Adobe file has been loaded, print a copy of the software operating manual.

Preparing HandiLaz Mini for Linking with PC:

1. Navigate from Options Menu to Utilities Menu. Set ID to a number from 0–31.
2. Navigate from Data Processing Menu to Uplink. Leave unit in Uplink Mode.
3. Connect the HandiLaz Mini to the PC via provided RS-232 cable.

Using PC Link Software

1. Start the software by going into the PMS directory (C:\Program Files\PMS) and double-clicking on the file entitled "Handheld LPC Data Retrieval Software."
2. Set the address to match the ID number (from 1–31) set on the HandiLaz Mini.
3. Confirm the Port is set for COM1.
4. Select the tab for the sampling mode of the data of interest (e.g., Repeat Mode).
5. Select "Read Data list" to load available data from the selected mode.

6 Review the data of interest as displayed on the screen.

Notes

- To see additional data on a record, highlight the record, then review the “Details” box. For example, the “Data list’ information will only tell you if there was an error, but the “Details” box will have information on exactly which type of error occurred (e.g., Laser Error or Flow Error).
- Select “Print All Records” to print using the PC or network printer.
- Select “Save All Records” to download data onto a CSV file saved on the PC.
- To read the time setting in the particle counter, or to reset the time in the particle counter, select the “Time Setup with HandiLaz” tab.

PC Requirements

CPU	Compatible IBM PC with greater than Pentium capacity.
Memory	>128 MB
Hard Disk Space	>50 MB when installed
Operating System	Microsoft: Windows 2000 Professional, XP Home Edition, XP Professional
Display	>800 $\times$ 600 resolution, with 32 true colors
Interface	9-pin COM port or USB 1.1 interface*

*If USB is used, a USB-RS232c converter required.

Communications Requirements

COM Port	COM1---COM8
Baud Rate	600
Data Bit	On
Stop Bit	1(fixed)

Troubleshooting Data Download

- Error Message: "*RS232c communications error!*" -- Verify the cable is properly connected, the handheld unit is set for UPLINK, and the PC Link software is connected to the same Address number as the ID number in the unit.
- Error Message: "*No Data!*" -- Ensure that the time period settings (i.e, from "Start Time" to "Stop Time") are set widely enough to include the time at which the samples of interest were taken. If you do change the time range, you must click "Data List" afterward to reset the list of data records to be displayed.

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Appendix A
Menu Tree

<MENU>
MEASURE MODE
DATA PROCESSING
OPTIONS

<MEASURE MODE> (Sampling Mode)
REPEAT (Take multiple samples)
SINGLE (Take one sample)
CONT (Sample Continuously until press STOP)
CALC (Take samples, make calculations)
REMOTE ***NOT SUPPORTED***
ISO (Take samples, perform ISO calculations)

<ISO END> (Completed sampling at this location)
NEXT POINT? (Start sampling at next location)
ERASE DATA? (Erase data from this location)
CALCULATE? (Calculate mean, etc. for locations sampled)
EDIT TIMES? (Edit # samples at next location)
FINISH? (Make final calculations; Save if STR=YES)

<DATA PROCESSING>
DISPLAY (Review stored data records)
UPLINK (Dedicate unit to communication with PC)
PRINT (Print designated records on external printer)
CLEAR (Erase memory)

<OPTIONS>
ALARM (Set alarm values for 0.3, 0.5, or 5.0 μm channels)
UNITS (CNT, /cf, or m3)
UTILITIES
HOTKEY
AUTOST (AUTOSTART)

<UTILITIES>
DATE (yyyy/mm/dd)
TIME
ID (0-31)
COMMUNICATION (RS232C, RS485)

<AUTOSTART>
DELAY (10-99 SEC)
AUTOSTART (ON, OFF)

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Appendix B
Abbreviations and Definitions

Term	Where Found	Meaning
ADDRESS	Screen	ID of handheld unit (0–31); used to link to PC
AUTOST.	Screen	AUTOSTART: Set Initial DELAY Set Automatic start
AVG/AVE/ MEAN/M	Screen, Data Retrieval Software	AVERAGE
B	Screen	BEEP enabled
CALC	Screen, Printout	CALCULATION Mode – Samples, then Calculates AVG, SD, MAX, and MIN
CALCULATE?	Iso Screen	Calculate ISO results for samples taken so far
CONT	Screen, Printout	CONTINUOUS Mode – Samples continuously until manually stopped.
CNT	Screen, Printout	COUNTS of particles
DELAY	Screen	DELAY before initial sample
E+2, E+5, etc.	Screen, Printout	Scientific Notation: *E+5 means: multiply the number to the left of "E" by 10 to the 5th power.*
E=	Screen, Printout	ERROR =
F	Screen, Printout	Error: Low FLOW
L	Screen, Printout	Error: LASER
O	Screen, Printout	Error: OVER Max. Concentration
EDIT TIMES?	Iso Screen	Change number of samples at next location
END	Screen	Last record to be printed
ENTER	Screen, Key	ENTER current command or parameter value

Term	Where Found	Meaning
FIN/FINISH	Screen	FINISHED sampling or subsampling
FREE REC.	Screen	Free space to store additional records
HOTKEY	Screen	Key that allows quick sampling start
IMPERF. TEST	Screen	Imperfect Test = Invalid sample
INT	Screen	Cycle INTERVAL = a) Time to conduct subsample, plus b) Delay before next subsample
ISO	Screen, Printout	ISO Mode: Used to certify that rooms meet ISO standards
LOC	Screen, Printout	LOCATION where sampling being performed
N/NO	Screen	NO; Not ready to sample
NEXT POINT/	Iso Screen	Start sampling at next location
OK	Screen	Ready to sample; All settings completed
POINT	Screen	LOCATION where sample/subsample taken (0–255)
PTCL	Screen	PARTICLE Display Units: “/M^3”, “/Ft.^3”, or CNT (i.e., Counts)
PREV	Screen, Printout	PREVIOUS screen
READY	Screen	Press START to begin sampling
REMOTE	Screen, Printout	NOTE: *This mode is not supported*
REPEAT	Screen, Printout	REPEAT Mode: Takes a series of identical samples (as defined in sampling settings)
RUN	Screen	Sampling underway
SAMPLE mm:ss	Screen	Duration of single sample in minutes and seconds
SAMPLETIME	Screen	Total time sampling at location.
SD	Screen	Standard Deviation
SINGLE	Screen, Printout	SINGLE Sample Mode: Takes a single sample as defined in sampling settings)
START	Screen	First record to be Displayed or Printed

Term	Where Found	Meaning
STR	Screen	STORE = Saves data inside HandiLaz Mini buffer
T/TIMES	Screen	TIMES one location is sampled
TOTAL TIME	Screen	Total time sampling and inter-sample delay at location
UCL	Screen	Upper Confidence Limit of 95% (used in ISO calculations)
UNIT	Screen	Set data UNITS (i.e., CNTS, Cnts/Ft.^{3} or Cnts/M^3)
UPLINK/LINK	Screen	Connects HandiLaz Mini to PC; PC can then copy sample records from HandiLaz Mini
UTILITY	Screen	Set Time, Date, Address & Comm Format
wait	Screen	Unit not yet prepared to sample
Y YES	Screen	YES; Ready to sample
1/3, 2/5, 4/4, etc.	Screen	Number of subsample / Total number of subsamples at this location

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Appendix C
Printer Setup

The HandiLaz Mini can directly produce print output by means of an optional printer and print cable. Before the printer is used for the first time, its Mode settings and Serial settings must be set to be compatible with HandiLaz Mini.

Setting the Printer's Functions

Printer function is set by responding to printed questions with the printer's control keys. FEED is yes and CHARGE is no.

To set the printer's function:

1. Ensure that the printer is off and connected to power. The power switch must be in the **OFF** position.
2. While pressing and holding down the **FEED** and **CHARGE** switches, move the power switch to the **ON** position.

   This puts the printer into the Function Set Mode.

   The printer will print "FUNCTION SETTING MODE" followed by the default factory settings.
3. After the printer prints ENTER MODE SETTING [YES/NO], press FEED (yes).

   COMMAND MODE = x will be printed.
4. If COMMAND MODE = A, press CHARGE (no).

   The printer will respond with COMMAND MODE = B.
5. Press FEED (yes), meaning "Yes, I want COMMAND MODE set to B."

   The printer will respond with the next setting, CHARACTER SET = xxx.

6 As each of the Function Setting Mode statements are printed, respond with the **FEED** or **CHARGE** switch to achieve the following settings:

ENTER MODE SETTING[YES/NO]			Press **FEED** (yes)
COMMAND MODE	**=**	**MODE B**	
CHARACTER SET	**=**	**GRAPHICS**	
ENTER SERIAL SETTING[YES/NO]			Press **FEED** (yes)
BIT LENGTH	**=**	**8BIT**	
PARITY	**=**	**NONE**	
FLOW CONTROL	**=**	**RTS/CTS**	
BAUD RATE	**=**	**9600bps**	

About the BAUD RATE: Continue pressing CHARGE (no) until **BAUD RATE = 9600** prints. Then, press **FEED** (yes)

After **FEED** (yes) is pressed in response to the print line **BAUD RATE = 9600**, the printer will respond with **SETTING COMPLETED**.

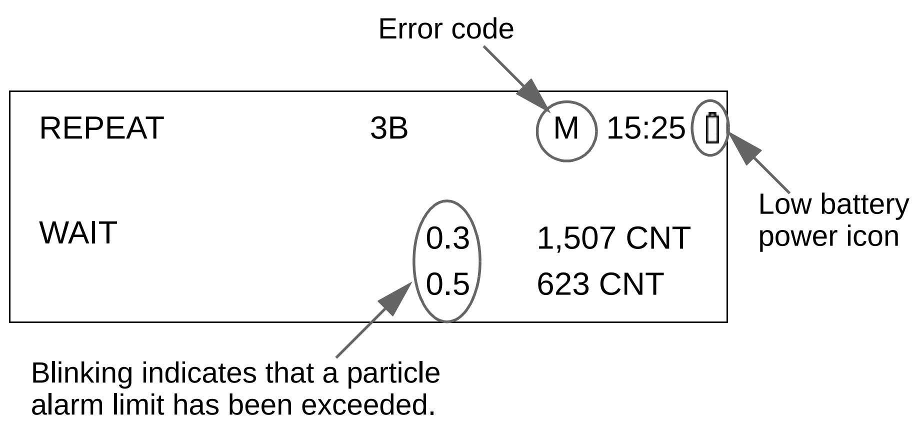
The printer is ready for use.

Appendix D
Error Messages and Alarms

When sampling, HandiLaz Mini monitors its operation, available power, and alarm settings. When an error or warning condition occurs, the display changes to alert the user. Indications of an error appear in the following forms:

- An Error code blinks in the form of blinking letters (L, F, O, or M) to the left of the time display.
- A low battery power symbol appears to the right of the time display.
- The particle size label(s) blinks to indicate that a particle limit setting has been exceeded.

Error Code and Alarm Locations



Error Code and Symbol Legend

Error Code or Symbol	**Error Type**	**Error explanation and recommended response**
L	Laser power abnormal	Stop sampling and, if problem persists, contact your Particle Measuring Systems representative.
F	Flow rate error	The flow of sample through the HandiLaz Mini is out-of tolerance (2.83 L/min. ±5%). If a filter and/or connecting tube are attached to the inlet, remove them. Check to see if sample inlet or tubing is obstructed. If the “F” symbol persists, contact your Particle Measuring Systems representative.
O	Maximum measurable concentration	The sample’s particle concentration is greater than allowable for the HandiLaz Mini. Stop sampling. Run unit with zero filter to clean sample chamber.
M	Printer buffer full	Turn printer OFF, the repeat normal printer operation.
▯	Low battery power	The remaining battery power is less than 4.1 V. Immediately replace the batteries with fully charged batteries, or start operating HandiLaz Mini with its power convertor.

Error Code Priority

Error codes are prioritized to display the most critical condition first. The priority in order of importance is: L, F, and O.

Low Battery Power

The indication of low battery power appears in two stages. When the batteries have discharged to a level less than 4.1 Volts, the first indicator appears as an empty battery icon. ▯

After a few minutes, the HandiLaz Mini will stop sampling and saving data.

Appendix E
Data Retrieval Software, Installing and Running

This software enables data to be downloaded from the HandiLaz Mini to a PC.

Minimum System Requirements

The following minimum system requirements must be provided for the software:

CPU	IBM Compatible PC, Pentium
Communications interface	• 9-pin COM port or • USB 1.1 **NOTE: Using the USB interface requires a USB RS-232C converter**
Operating system	• Microsoft Windows 2000 Professional, • Microsoft Windows XP Home Edition, or • Microsoft Windows XP Professional
Memory (RAM)	≥ 128 MB
Hard drive free space	≥ 50 MB
Display	• Resolution: 800 × 600 • Colors: 32 true colors

Software Installation

We recommend that other application programs be closed before installing this software.

NOTE: If a previous version of HandiLaz Mini software in loaded on the PC, errors may occur when this new edition is installed. Un-install the previous version before installing this software.

Insert the installation CD into the CD-ROM drive. The “welcome” dialog box will open. Click on the **OK** button.

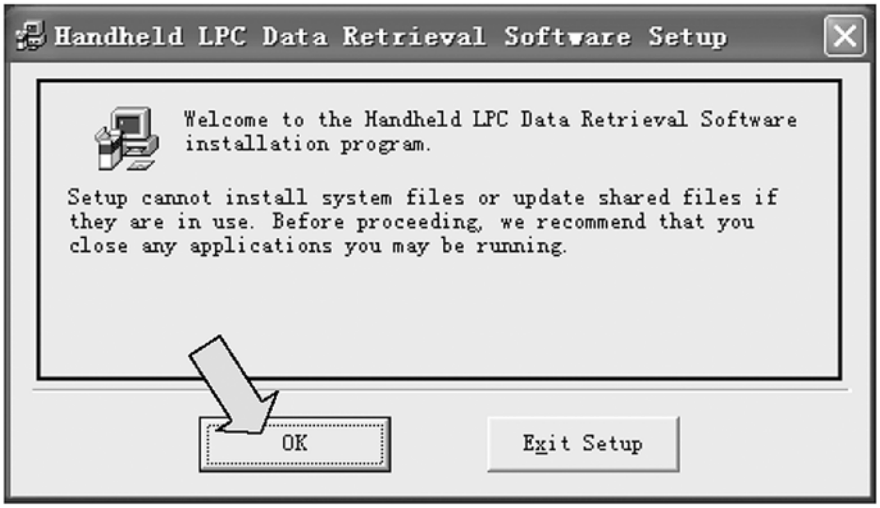


Figure E-1: Welcome Dialog Box

If after a few seconds the installation software does not start up automatically, use the following steps to open the .exe file manually:

1. Using Explorer®, find the CD drive and open it.
2. Find the Setup.exe file in the installation directory.
3. Double-click the Setup.exe file.

The Directory Selection dialog box will open.

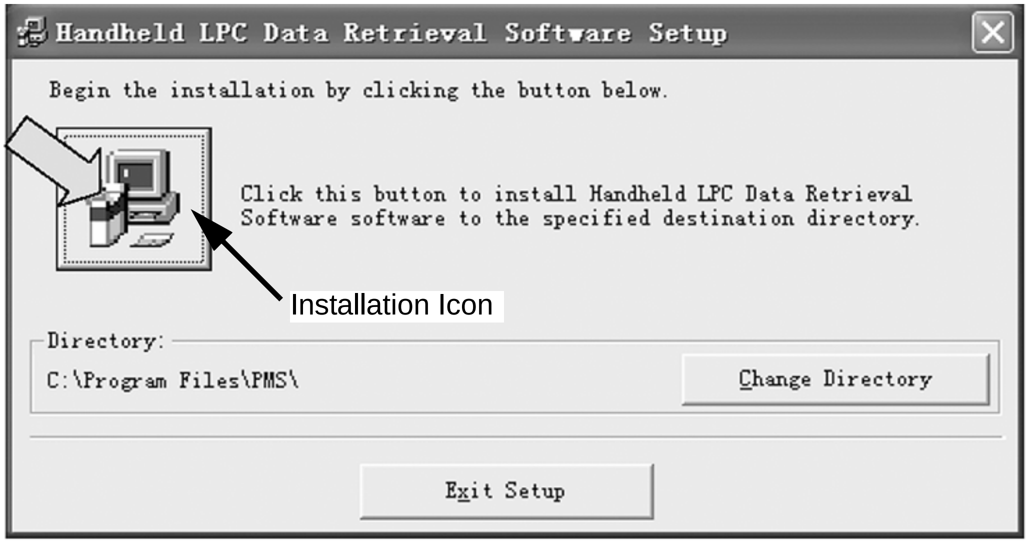


Figure E-2: Start Installation

Use the default directory, or click on **Change Directory** to choose the folder that will contain the software.

Click on the installation icon.

Choose the Program Group for software installation. The installation program defaults to the group "PMS".

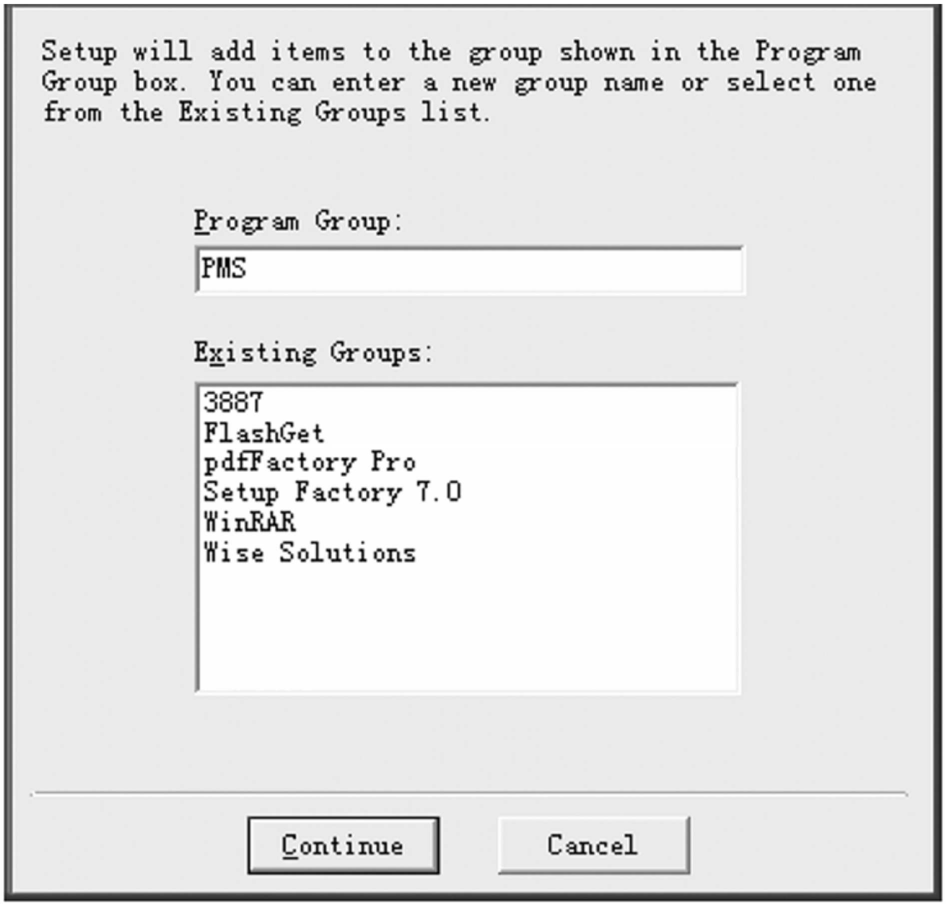


Figure E-3: Program Group Box

Click on the **Continue** button, when the correct Program Group has been selected.

The progress dialog box will appear, and after a few seconds the software will begin loading.

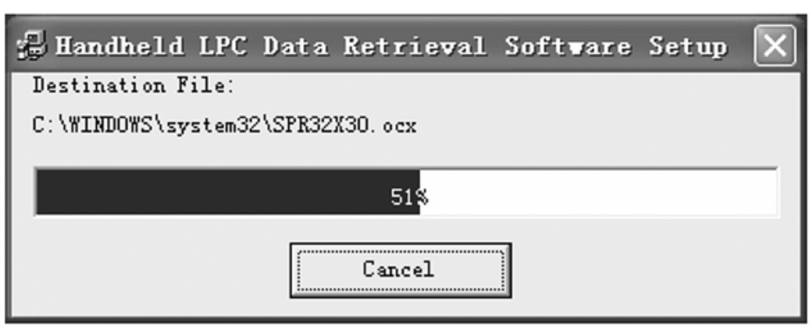


Figure E-4: Installation Progress

After the software is installed, the following dialog box will appear. Click on the **OK** button.

Figure E-5: Installation Successful

Using the Retrieval Software

HandiLaz Mini must be physically connected to the PC on which the retrieval software is loaded.

Connect the PC to the HandiLaz Mini

Use the following figures to connect HandiLaz Mini to the PC:

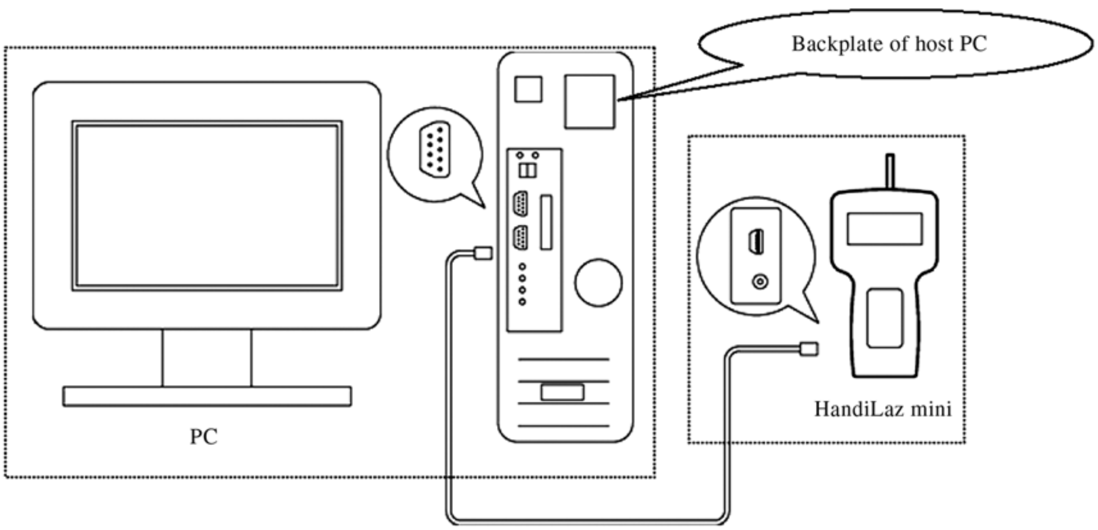


Figure E-6: Connection by means of RS-232 Cable

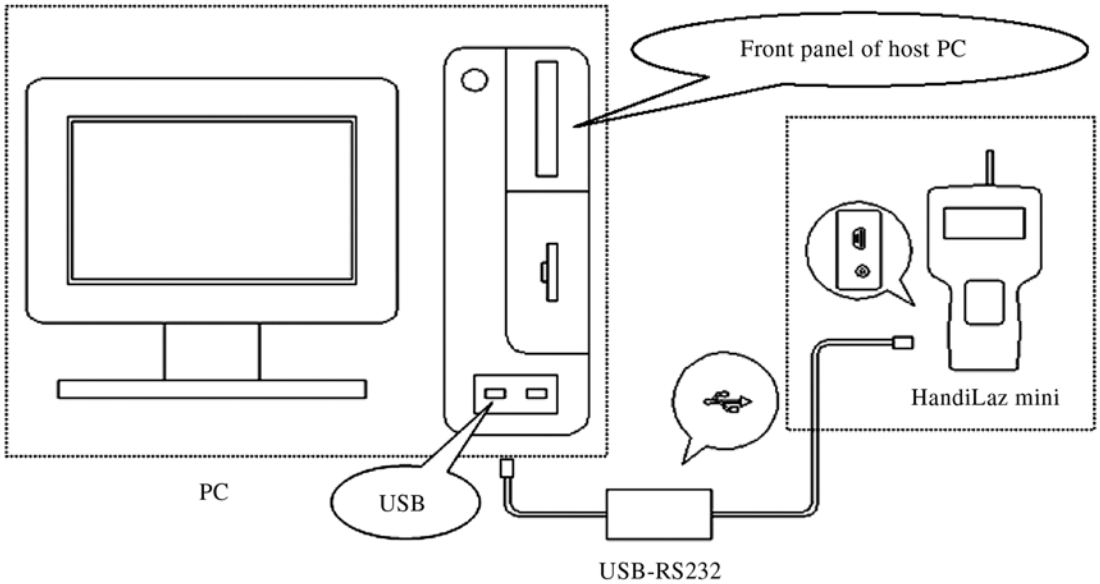


Figure E-7: Connection by means of USB Cable

Start the Retrieval Software

1. Click on the Windows **Start** button at the bottom left of the monitor.
2. Click on Programs > PMS-Handheld LPC PC-LINK. The program will start, and the Retrieval Software dialog box will appear on the monitor.

Start HandiLaz Mini and Verify or Set its Address

1. Press the **POWER** key to start the HandiLaz Mini. The <MENU> display will appear on the HandiLaz Mini.
2. From the MENU display use the up ▲ or down ▼ key to navigate to OPTIONS.
3. Press the **ENTER** button. The <OPTIONS> menu will open.
4. Navigate to ADDRESS.
5. If the address has already been set, record it and proceed to the next set of instructions.
   If the address has not been set, use the up ▲ or down ▼ key to set the two-digit address field to a number from 00 to 31.
6. Record the address and proceed to the next set of instructions.

Set HandiLaz Mini to Upload Data

1. From the MENU display use the up ▲ or down ▼ key to navigate to DATA PROCESSING.

2 Press the **ENTER** button. The <DATA PROCESSING> menu will open.

3 Navigate to UPLINK and press the **ENTER** key. HandiLaz Mini is now ready to communicate with a PC by means of a communication cable and the HandiLaz Mini Data Retrieval software.

Upload Data to the PC

HandiLaz Mini Data Retrieval Software makes it possible to download (save) data to a PC that has the retrieval software.

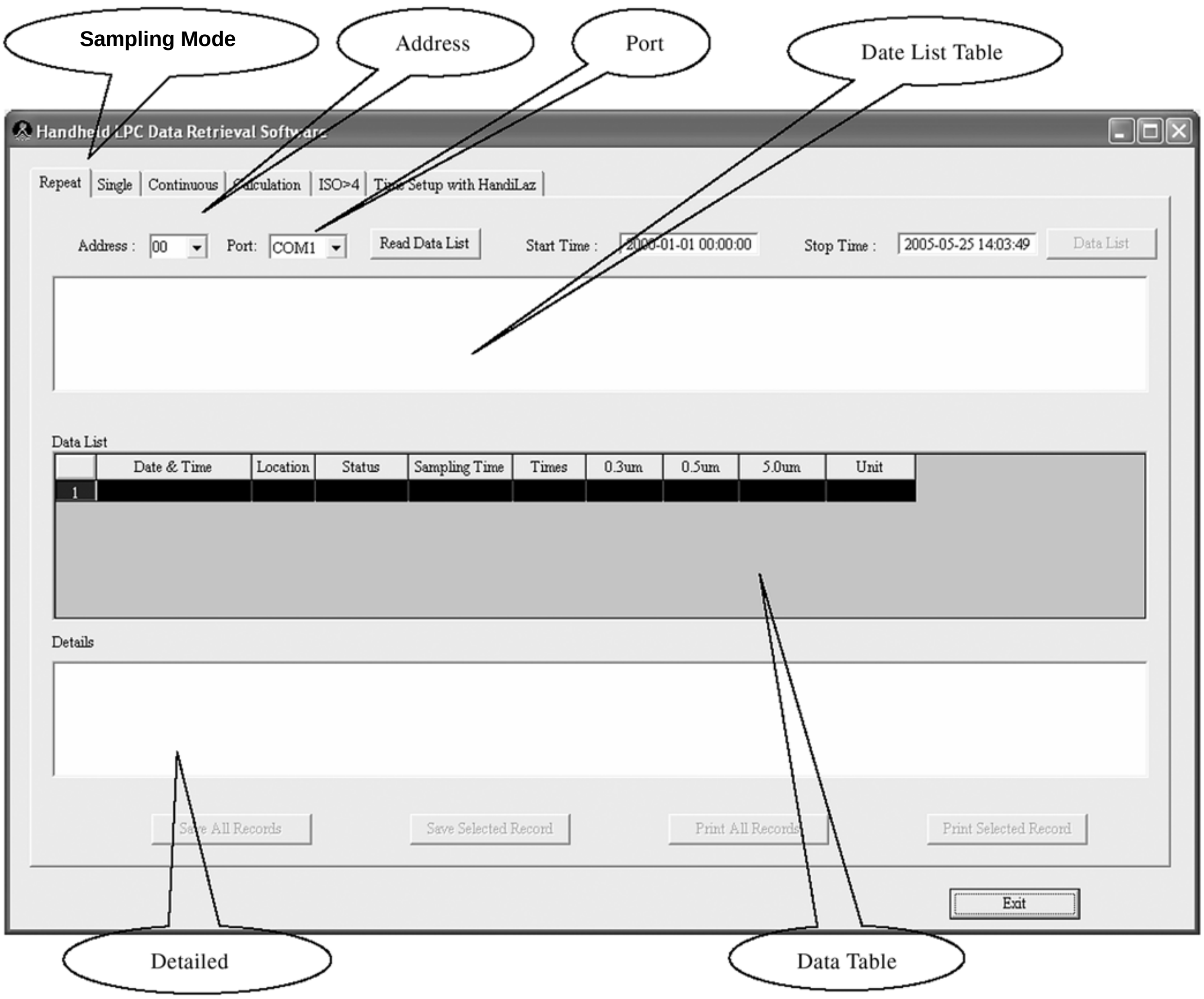


Figure E-8: Data Retrieval Software Main Dialog Box

Select the Sampling Mode of the Data to be Uploaded

1. At the top of the Data Retrieval Software, click on the appropriate tab for the mode (Repeat, Single, etc.) to be displayed.
2. Type the HandiLaz address (00–31) in the **Address:** field.
3. Set the **Port:** field to COM1–COM8. This is the COM port on the PC that the communications cable is connected to.
4. Click the **Read Data List** button. All the data stored for the selected mode will appear in the Date List Table and the Data Table. See Figure 2-4.

Figure E-9: Data Display from the Repeat Mode

Error Messages

If there is a problem with communications or data, one of the following error messages will appear:

If this RS 232 message appears, check the cable connections.

If this message appears, no data exists for the selected sampling mode. Try selecting another sampling mode.

Save all Data

1 Click the **Save All Records** button. The **Save Data** dialog box will open.

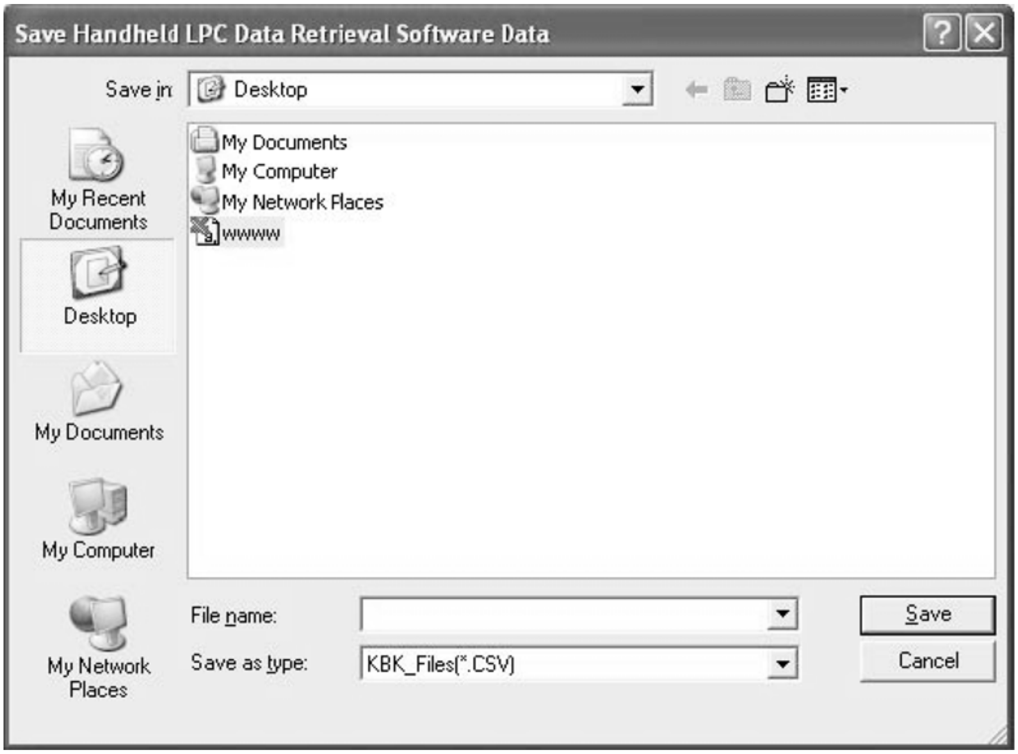


Figure E-10: File Location, Name, and Extension

2 In the **Save as type:** field, select the **CSV** file type.

3 In the **File name:** field, change the file name as needed.

4 In the **Save in:** field, select the location the data will be saved to.

5 When all fields are correct, click the **Save** button. The data will be uploaded to the PC as specified.

Save Data from a Time/Date Range

1 On the upper-right part of the Data Retrieval Software display, type the **Start Time:** (date and time).

2 Type the **Stop Time:** (date and time).

3 Click on the **Data List** button. Only the data within the specified dates/times will be displayed.

4 Click the **Save Selected Records** button. The Save Data dialog box will open.

5 In the **Save as type:** field, select the **CSV** file type.

6 In the **File name:** field, change the file name as needed.

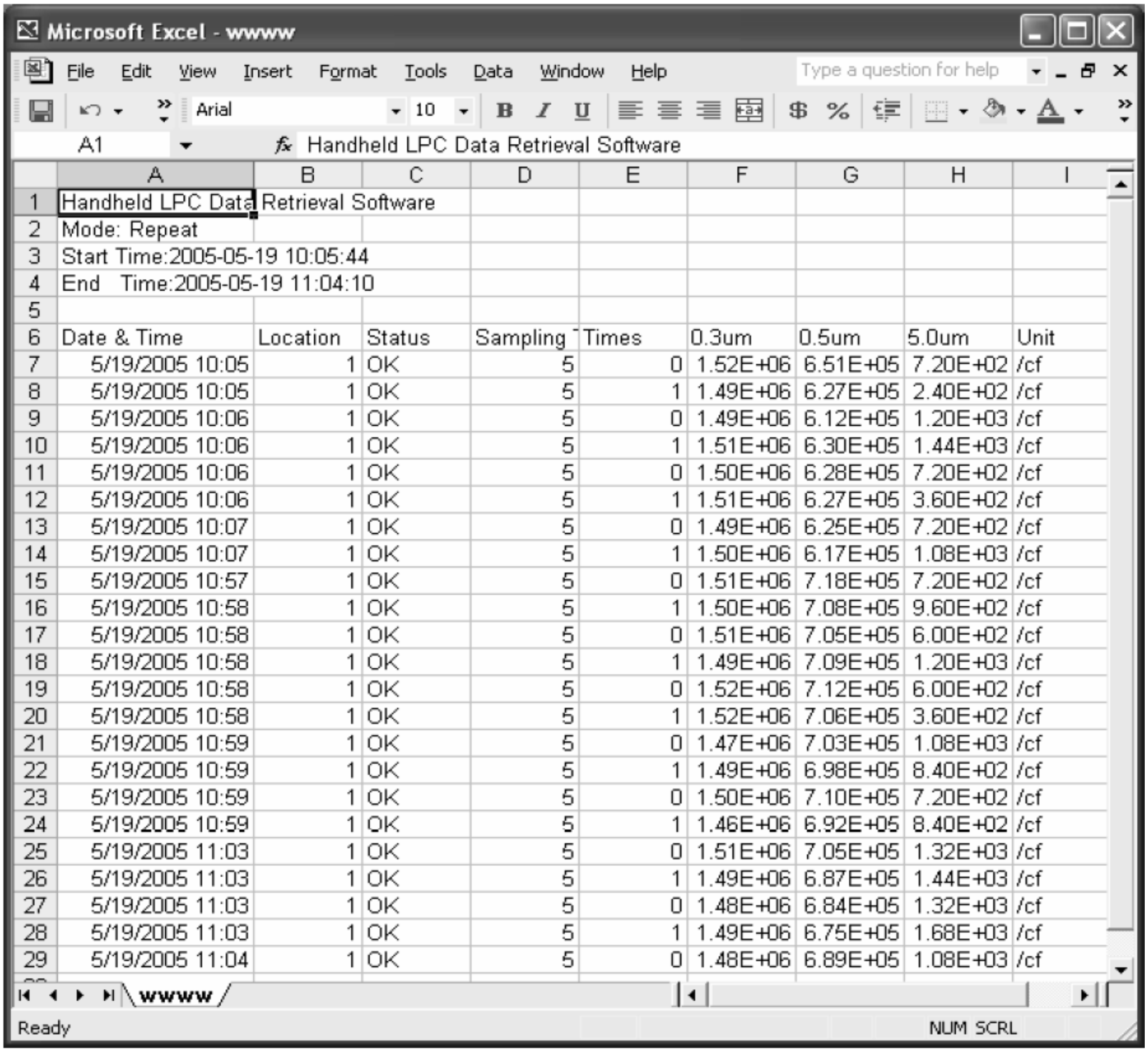
7 In the **Save in:** field, select the location the data will be saved to.

8. When all fields are correct, click the **Save** button. The data will be uploaded to the PC as specified.

View the CSV File with Microsoft Excel

To view the CSV file in Excel:

1. Open Excel.
2. Choose **Open**.
3. In the dialog box, find **File of type** and choose **All files**.
4. Find your file and select **Open**. The text Import Wizard will open.
5. Choose **Delimited** and **NEXT**.
6. Choose **Comma** and **Finish**.



	A	B	C	D	E	F	G	H	I
1	Handheld LPC Data Retrieval Software								
2	Mode: Repeat								
3	Start Time:2005-05-19 10:05:44								
4	End Time:2005-05-19 11:04:10								
5									
6	Date & Time	Location	Status	Sampling T	Times	0.3um	0.5um	5.0um	Unit
7	5/19/2005 10:05	1	OK	5	0	1.52E+06	6.51E+05	7.20E+02	/cf
8	5/19/2005 10:05	1	OK	5	1	1.49E+06	6.27E+05	2.40E+02	/cf
9	5/19/2005 10:06	1	OK	5	0	1.49E+06	6.12E+05	1.20E+03	/cf
10	5/19/2005 10:06	1	OK	5	1	1.51E+06	6.30E+05	1.44E+03	/cf
11	5/19/2005 10:06	1	OK	5	0	1.50E+06	6.28E+05	7.20E+02	/cf
12	5/19/2005 10:06	1	OK	5	1	1.51E+06	6.27E+05	3.60E+02	/cf
13	5/19/2005 10:07	1	OK	5	0	1.49E+06	6.25E+05	7.20E+02	/cf
14	5/19/2005 10:07	1	OK	5	1	1.50E+06	6.17E+05	1.08E+03	/cf
15	5/19/2005 10:57	1	OK	5	0	1.51E+06	7.18E+05	7.20E+02	/cf
16	5/19/2005 10:58	1	OK	5	1	1.50E+06	7.08E+05	9.60E+02	/cf
17	5/19/2005 10:58	1	OK	5	0	1.51E+06	7.05E+05	6.00E+02	/cf
18	5/19/2005 10:58	1	OK	5	1	1.49E+06	7.09E+05	1.20E+03	/cf
19	5/19/2005 10:58	1	OK	5	0	1.52E+06	7.12E+05	6.00E+02	/cf
20	5/19/2005 10:58	1	OK	5	1	1.52E+06	7.06E+05	3.60E+02	/cf
21	5/19/2005 10:59	1	OK	5	0	1.47E+06	7.03E+05	1.08E+03	/cf
22	5/19/2005 10:59	1	OK	5	1	1.49E+06	6.98E+05	8.40E+02	/cf
23	5/19/2005 10:59	1	OK	5	0	1.50E+06	7.10E+05	7.20E+02	/cf
24	5/19/2005 10:59	1	OK	5	1	1.46E+06	6.92E+05	8.40E+02	/cf
25	5/19/2005 11:03	1	OK	5	0	1.51E+06	7.05E+05	1.32E+03	/cf
26	5/19/2005 11:03	1	OK	5	1	1.49E+06	6.87E+05	1.44E+03	/cf
27	5/19/2005 11:03	1	OK	5	0	1.48E+06	6.84E+05	1.32E+03	/cf
28	5/19/2005 11:03	1	OK	5	1	1.49E+06	6.75E+05	1.68E+03	/cf
29	5/19/2005 11:04	1	OK	5	0	1.48E+06	6.89E+05	1.08E+03	/cf

Print all Records

1 Ensure that the PC and HandiLaz are running, connected appropriately, and that the Data Retrieval software is running.

1 From the Data Retrieval software dialog box, click on **Print All Records**.

2 Set the Printer dialog box as you would for any other print job.

3 Click on the **Print** button.

Print a Range of Records

1 On the upper-right part of the Data Retrieval Software display, type the **Start Time:** (date and time).

2 Type the **Stop Time**: (date and time).

3 Click on the **Data List** button. Only the data within the specified dates/times will be displayed in the Date List Table and the Data Table.

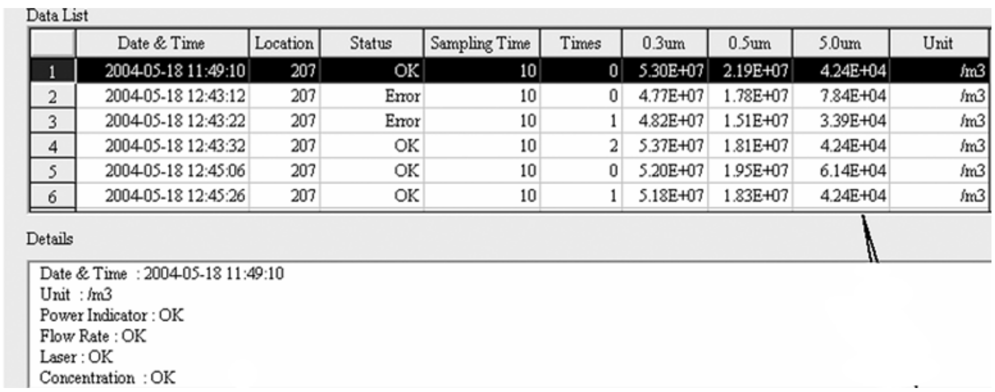
4 Click on the **Print Selected Records** button. The Windows Print dialog box will open.

5 Set the Printer dialog box as you would for any other print job.

6 Click on the **Print** button.

View the Details of a Data Record

To see more detailed information about a record, simply click on that record, and view the information in “Details”.



	Date & Time	Location	Status	Sampling Time	Times	0.3um	0.5um	5.0um	Unit
1	2004-05-18 11:49:10	207	OK	10	0	5.30E+07	2.19E+07	4.24E+04	/m3
2	2004-05-18 12:43:12	207	Error	10	0	4.77E+07	1.78E+07	7.84E+04	/m3
3	2004-05-18 12:43:22	207	Error	10	1	4.82E+07	1.51E+07	3.39E+04	/m3
4	2004-05-18 12:43:32	207	OK	10	2	5.37E+07	1.81E+07	4.24E+04	/m3
5	2004-05-18 12:45:06	207	OK	10	0	5.20E+07	1.95E+07	6.14E+04	/m3
6	2004-05-18 12:45:26	207	OK	10	1	5.18E+07	1.83E+07	4.24E+04	/m3

Change the Current Time on the HandiLaz Mini

1 Ensure that the HandiLaz Mini is connected to the PC.

2 At the top of the Data Retrieval software dialog box, click on **Time Setting with HandiLaz**. The time setting dialog box will open.

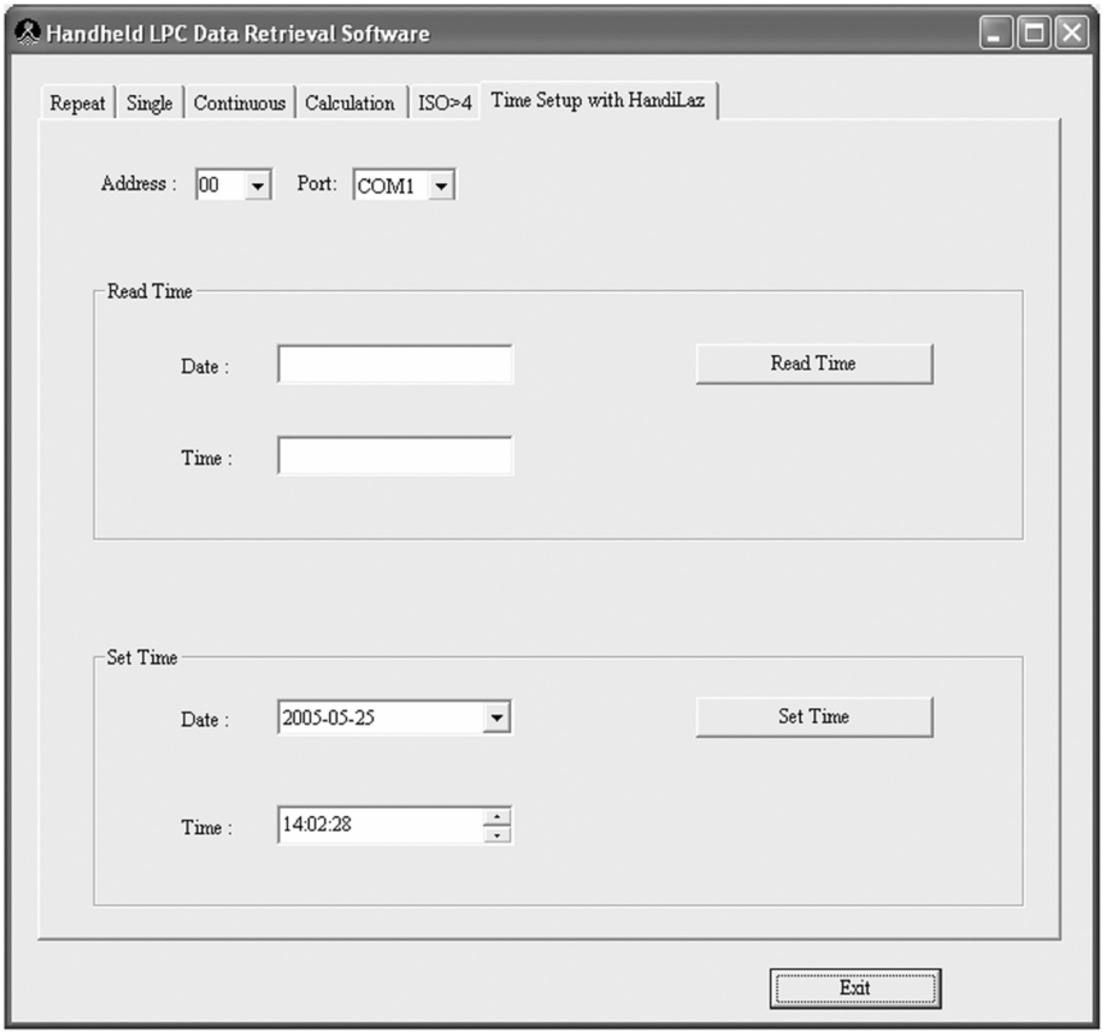


Figure E-11: Set HandiLaz Mini System Time from Data Retrieval Software

3 Click the **Read Time** button. The current time from the HandiLaz Mini will be read to the **Read Time**/**Date:** and **Time:** fields.

4 Type the new date and time into the **Set Time/Date:** and **Time:** fields respectively.

5 Click the **Set Time** button. The current time in the HandiLaz Mini will be changed to the new date and time settings.

6 Click on **Exit** when all information is correct.

Appendix F
International Precautions

WARNING

This instrument is a Class I laser product. Use of controls, or adjustment, or performance of procedures other than those specified here may result in hazardous radiation exposure.

AVERTISSEMENT

Cet instrument contient un laser de Classe 1. L'usage de moyens ou procédures, ou la pratique de certains réglages autres que ceux prévus dans ce manuel peuvent entrainer des risques d'exposition aux radiations laser.

WARNUNG

Das Gerat ist ein Laser der Klasse 1. Werden Einstellungen vorgenommen oder Verfahren eingesetzt, die nicht in dieser Bedienungsanleitung beschrieben sind, kann es zu gefahrlicher Strahlung kommen.

ATTENZIONE

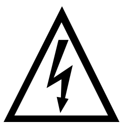
Questo strumento e nella classe 1 dei prodotti laser. Fare controlli o calibrazioni e seguire procedure diverse da quelle specificate in questo manuale, pu provocare esposizione a radiazioni pericolose.

PRECAUSION

Este equipo es un producto laser de Clase I. La utilización de controles, ajustes o procesos diferentes de los especificados aquí, pueden dar lugar a exposiciones a radiaciones peligrosas.

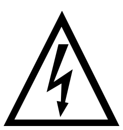
Hazard Symbols

The meaning of hazard symbols appearing on the equipment is as follows:

Symbol	Nature of Hazard
	Attention, consult accompanying documents.
	Dangerous High Voltage

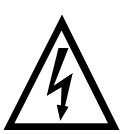
Symboles de risque

Des symboles représentant les risques sont placés sur l'appareil. Leur signification est la suivante:

Symbole	Nature du risque
	Attention, consulter les documents d’accompagnement
	Danger Electricite

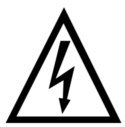
Warnschilder

Die, an dem Gerat angebrachten Warnschilder haben folgende Bedeutungen:

Symbol	Gefahrenart
	Achtung! In den beiliegenden Unterlagen nachschlagen
	Achtung Hochspannung

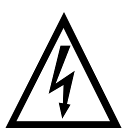
Simboli di pericolo

Il significato dei simboli di pericolo che appaiono sugli strumenti il seguente:

Simbolo	Natura del pericolo
	Attenzione. Consultare i documenti allegati
	Tensione Pericolosa

Simbolos de peligro

Los simbolos de peligro que aparecen en el equipo significan:

Símbolo	Naturaleza del Peligro
	Atención, consultar los documentos adjuntos.
	Peligro alto voltaje.

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